BEING UNIVERSALLY OPEN IS FPPF LOCAL ON THE SOURCE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file descent.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof refers to 'the Lemma above' when citing the locality of openness; the reference is kept as the author's own navigation stub.

Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Allowed standard prerequisites (editorial).

- An fppf morphism is universally open, and universal openness is preserved by composition. Morphisms, Lemmas morphisms-lemma-fppf-open and morphisms-lemma-composition-open, invoked at proof lines 7096-7100. Gives the forward direction.
- Openness is fppf local on the source. Same chapter, Lemma lemma-open-fppf-local-source, invoked at proof lines 7105-7107. Gives the converse direction, applied after base change to Y'.

Source reference. Original lemma 036P: sources/stacks-descent/pinned-original-source.tex, source lines 7066–7066. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 02V2: sources/stacks-chow/tags.txt, source lines 3753–3753. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 01UA: sources/stacks-chow/tags.txt, source lines 2403–2403. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *The property $\mathcal{P}(f) =$"f is universally open" is fppf local on the source.*

Proof. Let $f : X \to Y$ be a morphism of schemes. Let $\{X_i \to X\}_{i \in I}$ be an fppf covering. Denote $f_i : X_i \to X$ the compositions. We have to show that f is universally open if and only if each f_i is universally open. If f is universally open, then also each f_i is universally open since the maps $X_i \to X$ are universally open and compositions of universally open morphisms are universally open (Morphisms,

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Lemmas 01UA and 02V2). Conversely, assume each f_i is universally open. Let $Y' \to Y$ be a morphism of schemes. Denote $X' = Y' \times_Y X$ and $X'_i = Y' \times_Y X_i$. Note that $\{X'_i \to X'\}_{i \in I}$ is an fppf covering also. The morphisms $f'_i : X'_i \to Y'$ are open by assumption. Hence by the Lemma 036P above we conclude that $f' : X' \to Y'$ is open as desired. $\square$